CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

4.1 Turning Effects of Forces

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

What is the centre of gravity of an object?

A the geometrical centre of the object

B the point about which the total torque is zero

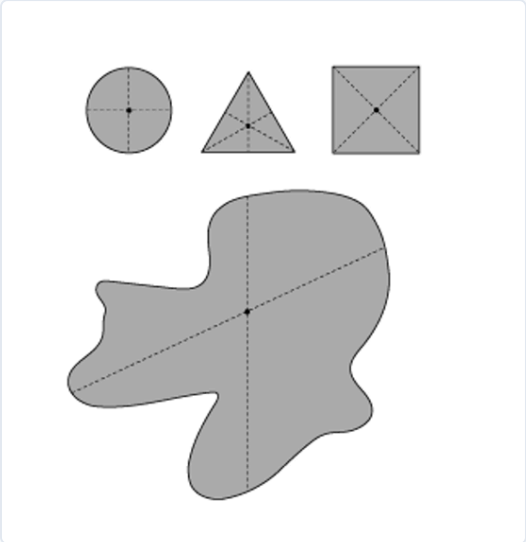
C the point at which the weight of the object may be considered to act

D the point through which gravity acts

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The centre of gravity of an object is defined as the point where all the weight of the object may be considered to act
- Here are some examples of the centre of gravity of some regular and irregular shapes:



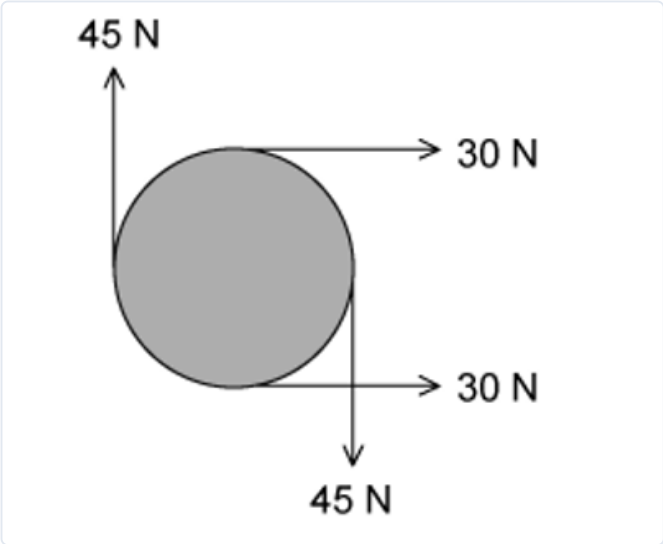
A is incorrect because this is true for regular shapes but not for irregular shapes

B is incorrect because the point about which the total torque is zero is the point the object is in rotational equilibrium

D is incorrect because the centre of gravity is where the **force of gravity** (weight) acts throughout the object

Question 2 Easy [1 mark]

The diagram shows four forces applied to a circular object.



Which of the following describes the resultant force and resultant torque on the object?

	Resultant force	Resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

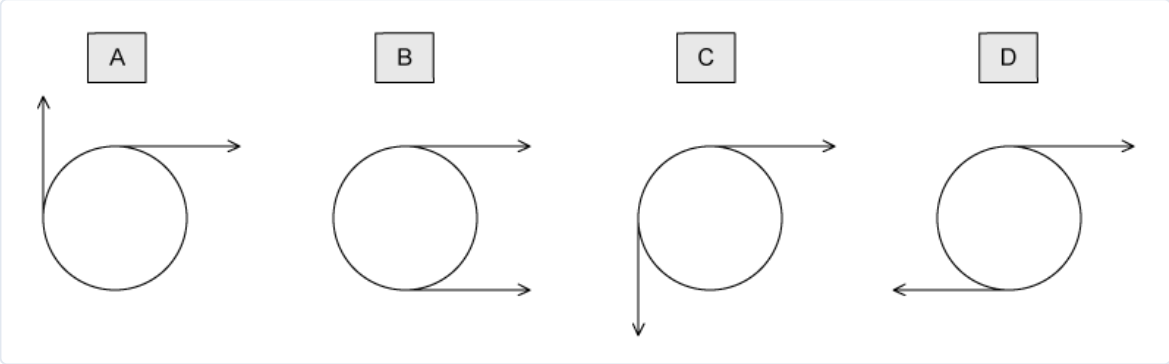
- Vertical resultant force = 45 – 45 = 0
- Horizontal resultant force = 30 + 30 = 60 N to the right
- Therefore, the resultant force is non-zero

- The two 45 N forces along with the upper 30 N force all cause a clockwise moment, although the lower 30 N causes an anticlockwise moment, overall, the resultant torque is non-zero

Question 3 Easy [1 mark]

Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude.

Which arrangement of forces provides only a couple?



A

B

C

D

Mark Scheme & Explanation:

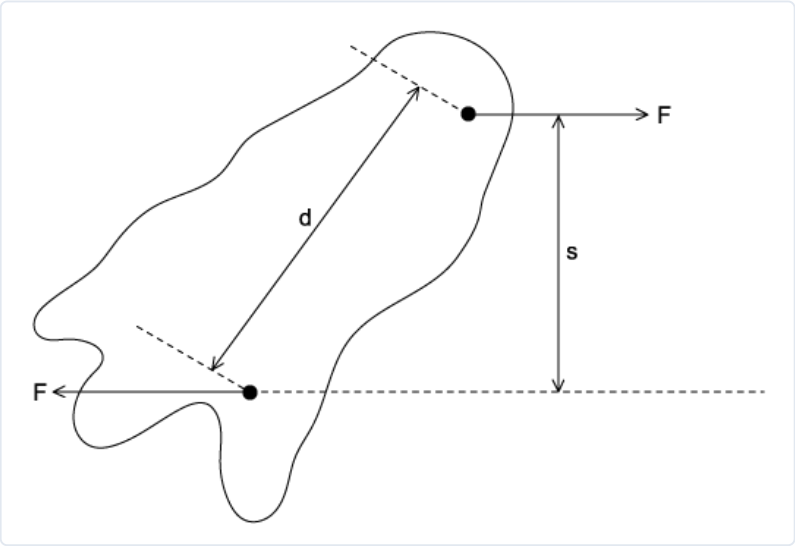
*The correct answer is **D** because:*

- The system of forces must provide a resultant torque, but we just need a couple so the resultant force should be zero as the forces should cancel out
- To form a couple, two forces must be equal in magnitude, parallel, but opposite in direction, and separated by a perpendicular distance
- Option D is the only one to satisfy this definition

A, B & C are incorrect because the directions of each of these force pairs do not oppose each other

Question 4 Easy [1 mark]

Two parallel forces, each of magnitude *F*, act on a body as shown



What is the magnitude of the torque on the body produced by this couple?

A *Fd*

B *Fs*

C 2 *Fd*

D 2*Fs*

Mark Scheme & Explanation:

*The correct answer is **B** because:*

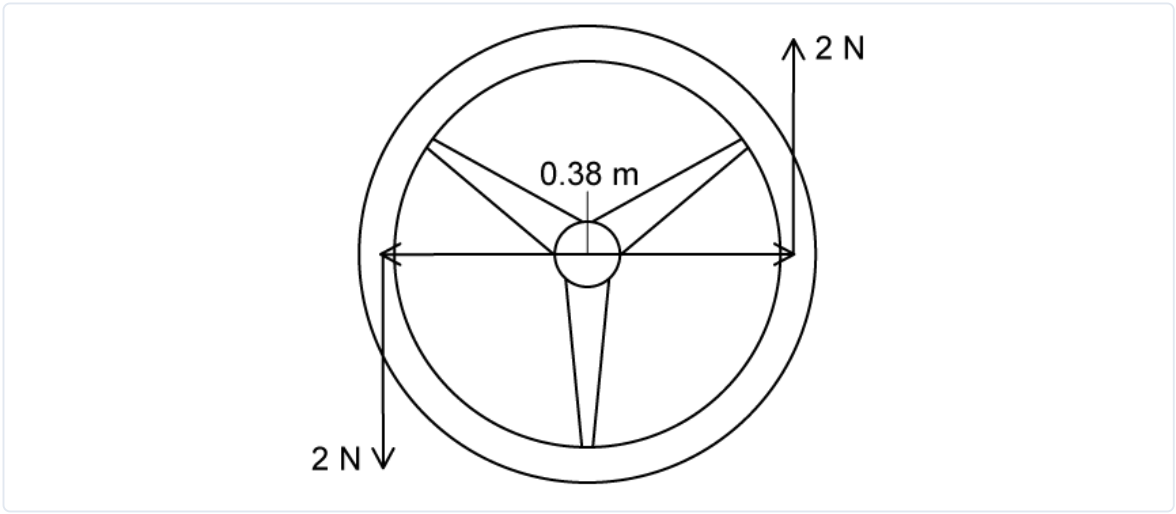
- A couple refers to two parallel forces that are equal in magnitude, parallel but opposite in direction and are at a **perpendicular** distance from each other
- The perpendicular distance is *s*
- So, the torque exerted by the couple is *Fs*

A is incorrect because the distance *d* is the distance between the forces, not the perpendicular distance

C & D are incorrect because the moment of a couple is calculated by multiplying the size of **one** of the forces by the perpendicular distance

Question 5 Easy [1 mark]

A steering wheel of diameter 0.38 m is rotated to the left by two forces of 2 N, as shown in the diagram.



What is the torque of the couple produced by the rotation of the steering wheel?

A 0.38 N m

B 0.50 N m

C 0.76 N m

D 1.0 N m

Mark Scheme & Explanation:

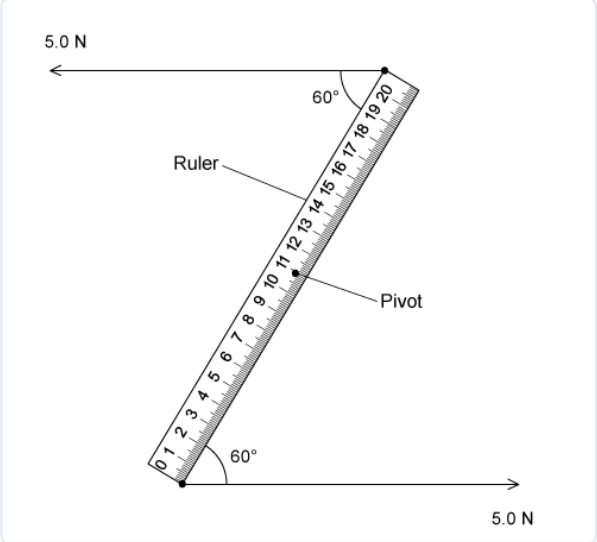
*The correct answer is **C** because:*

- The equation for the torque of a couple is given by:
  - Torque = Force × perpendicular distance between the lines of action of the forces
- Substituting in the numbers gives:
  - $\tau = 2 \times 0.38 = 0.76$ N m

Remember that the distance in the torque of a couple equation is the distance between both forces, **not** half of the distance!

Question 6 Medium [1 mark]

A 20 cm ruler is pivoted at its centre. Equal and opposite forces of magnitude 5.0 N are applied to the ends of the ruler, creating a couple as shown



What is the magnitude of the torque of the couple on the ruler when it is in the position shown?

A 0.2 N m

B 0.9 N m

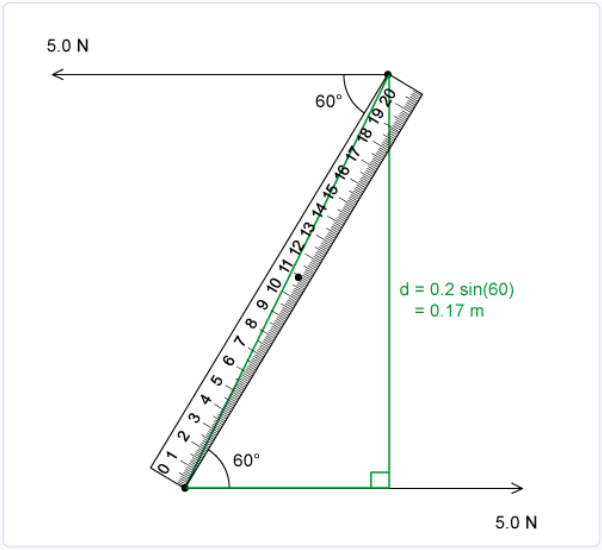
C 1.0 N m

D 1.7 N m

Mark Scheme & Explanation:

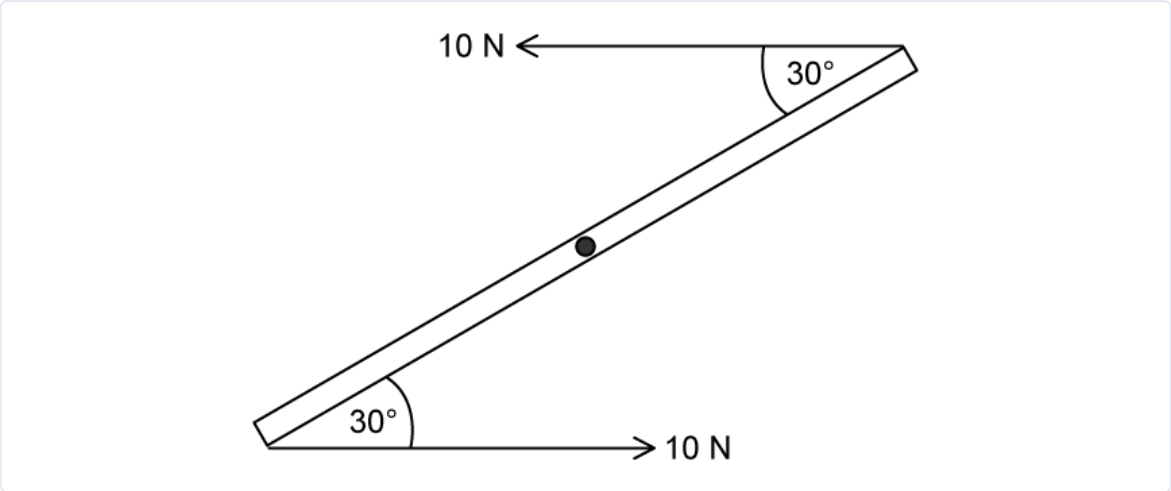
*The correct answer is **B** because:*

- A couple refers to two parallel forces that are equal in magnitude, parallel but opposite in direction and are at a **perpendicular** distance from each other
- The torque of a couple is found by calculating
  - $\text{torque} = \text{force} \times \text{perpendicular distance between forces}$
- Perpendicular distance between forces = $0.2 \sin 60°$
- Torque: $\tau = 5.0 \times 0.2 \sin 60° = 0.866 \approx 0.9 \text{ N m}$


5.0 N
60°
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
d = 0.2 sin(60)
= 0.17 m
60°
5.0 N

Question 7 Medium [1 mark]

A 10 cm beam is pivoted at its centre. Equal and opposite forces of 10 N pull the ends of the beam, creating a couple as shown.



What magnitude is the torque of the couple?

A 0.3 N m

B 0.5 N m

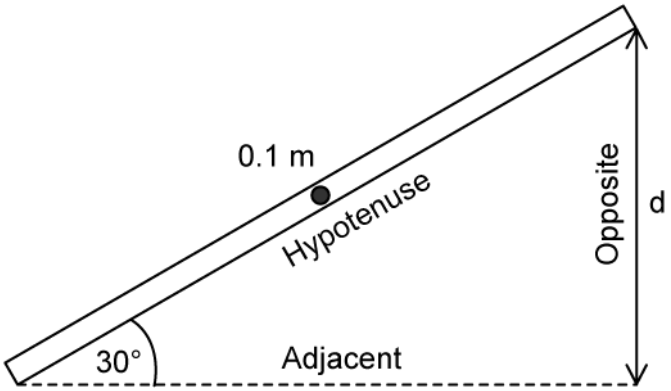
C 0.9 N m

D 1.0 N m

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- To calculate the torque of a couple:
  - $\text{torque}\ \tau = \text{force}\ F \times \text{distance between forces}\ d$
- The distance between forces is calculated using trigonometry:



- $\sin\ \theta\ =\ \frac{\text{opposite}}{\text{hypotenuse}}$
- $\sin\ 30°\ =\ \frac{d}{0.1}$
- $d\ =\ 0.1\ \sin\ 30°\ =\ 0.05\ \text{m}$
- Using this value for *d* in the torque equation:
  - $\tau\ =\ F\times\ d$
  - $\tau\ =\ 10\ \times\ 0.05\ =\ 0.5\ \text{N m}$

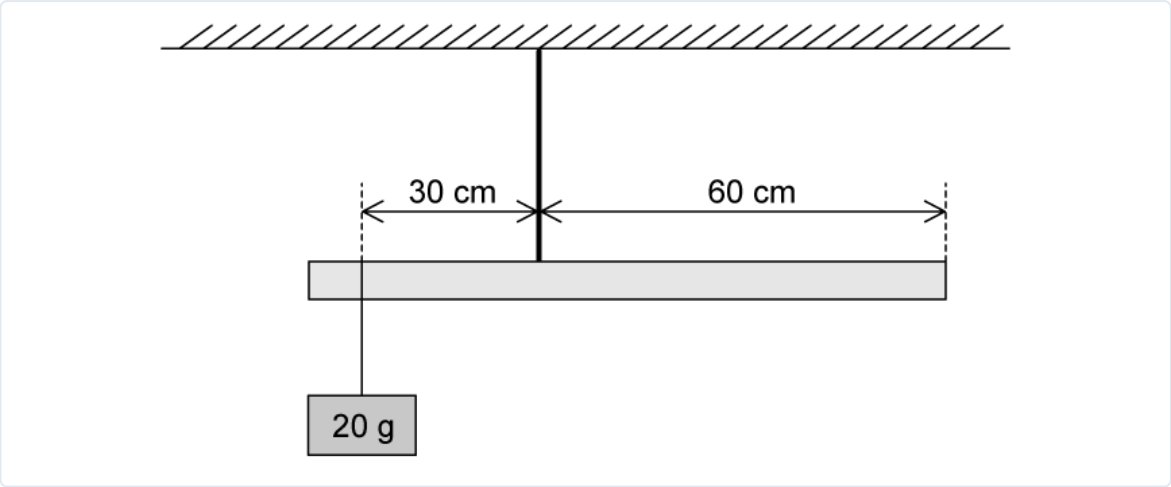
A is incorrect because this only takes into account the force on one side of the pivot.

C is incorrect because this is the answer if cosine is used instead of sine when calculating the perpendicular distance between the lines of action of the forces.

D is incorrect because the full length of the beam has been used instead of the perpendicular distance between the lines of action of the forces.

Question 8 Medium [1 mark]

A 1 m long uniform beam is hung from the ceiling by a rope. and is balanced when a 20 g mass is positioned 30 cm from the rope as shown.



What is the mass of the metre rule?

A 0.06 g

B 20 g

C 60 g

D 100 g

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- We will take moments about the pivot, i.e. where the rope is tied to the beam
- The anticlockwise moment of the 20 g mass, which is 30 cm from the pivot, is:
  - *moment* = *force* × *distance*
  - *moment* = 20 × *g* × 30 = 600*g*
  - Where *g* is the acceleration due to gravity
- To calculate the clockwise moment we need to know the distance of the centre of mass of the ruler from the pivot point
- As the ruler is uniform this will be at the halfway point of 50 cm
  - Therefore, the distance from the pivot point is 60 − 50 = 10 cm

- The clockwise moment due to the mass M of the beam is:
  - $moment = 10 \times M \times g = 10Mg$
- Using the principle of conservation of moments:
  - $10Mg = 600g$
  - $M = \frac{600}{10} = 60$ g
- This is option **C**

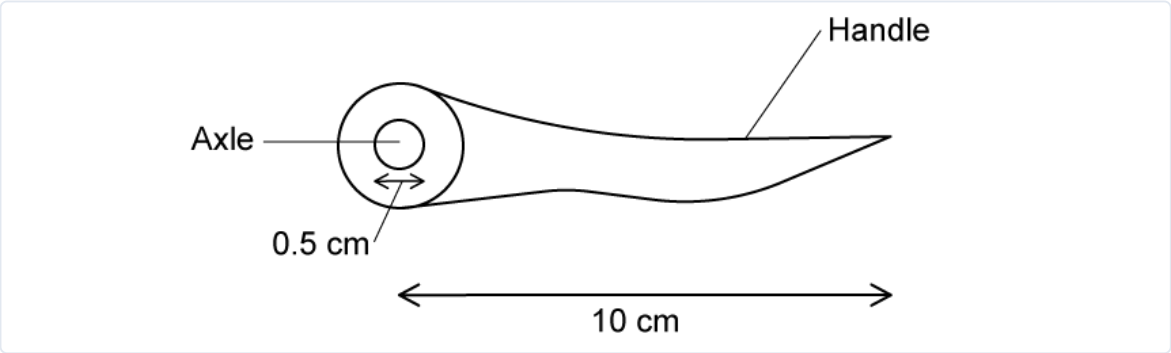
A is incorrect because the final answer has been converted into kg and not been converted back into grams

B is incorrect because a 30 cm from the pivot point distance has been used for the centre of mass of the beam

D is incorrect because the end of the beam has been used as its centre of mass

Question 9 Medium [1 mark]

A door handle in the diagram below has a length 10 cm requires a force of 20 N to turn it.



If the handle breaks and the door must be opened using the axle of the handle, with a diameter of 0.5 cm, how much force is needed?

A 4 N

B 8 N

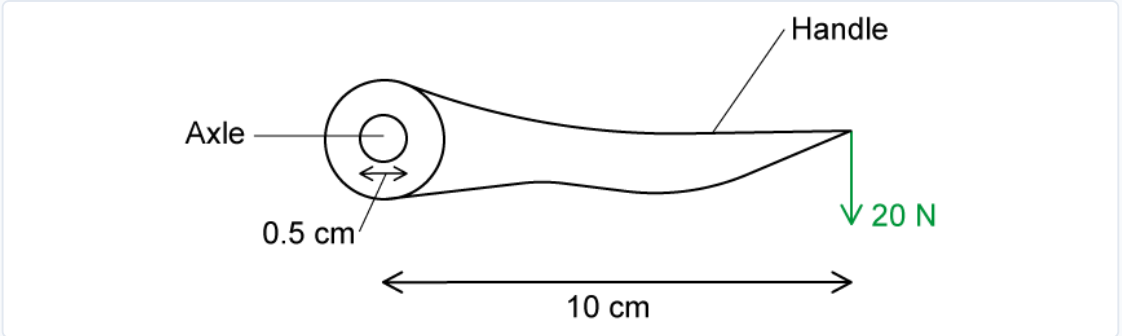
C 40 N

D 400 N

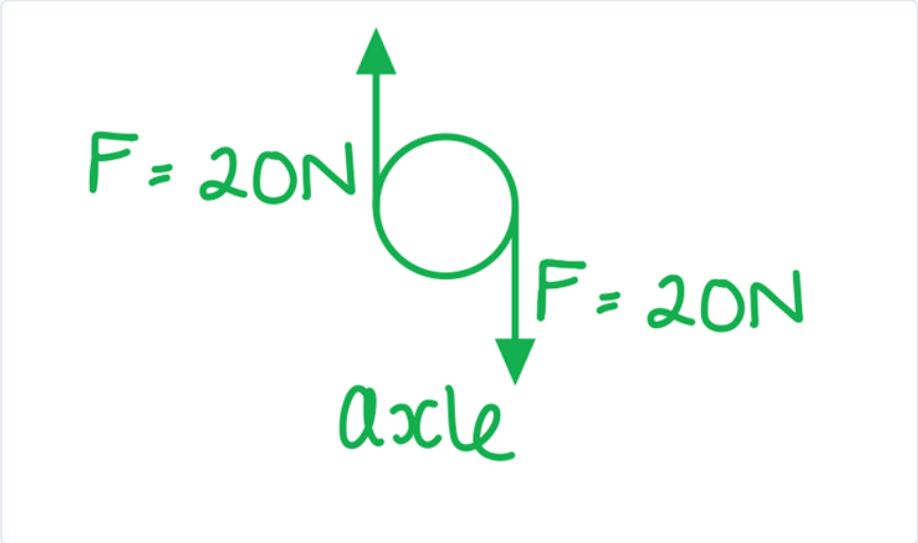
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The moment of a force is defined as:
  - Moment = Force × Perpendicular distance from the pivot
- The moment required when the handle is intact is:
  - Moment = 20 × 0.1 = 2 N m



- This is the same moment that's required by the axle in order to open the door
- Since the axle is circular, the turning force would produce a **couple**



- The torque of a couple is:
  - Torque = Force × Perpendicular distance between the line of action of the forces
- This means the force is equal to:
  - Force = $\frac{\text{torque}}{\text{perpendicular distance between forces}}$
  - Force = $\frac{2}{0.005}$ = 400 N

A is incorrect as the distance from the centre of the axle to the force causing the couple has not been converted from cm to m

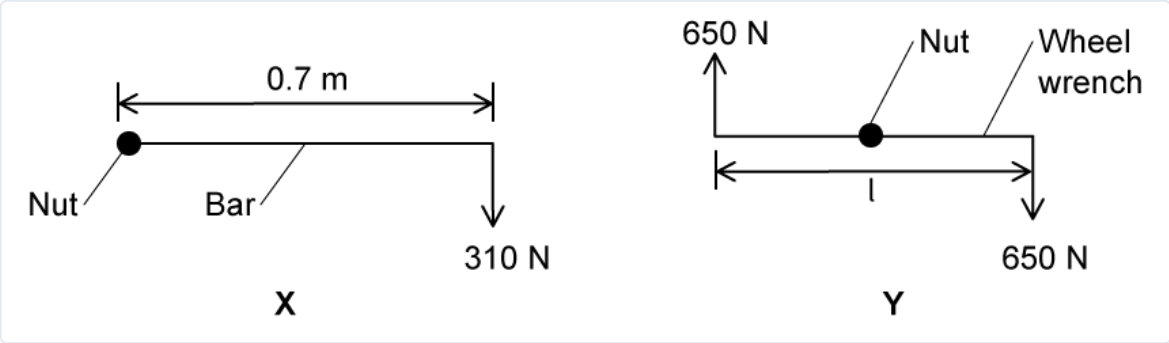
B is incorrect as the distance from the centre of the axle to the force causing the couple has not been converted from cm to m and 2 × 20 N is used instead of just 20 N

C is incorrect as 2 × 20 N is used instead of just 20 N (or 0.025 cm has been used as the distance instead of 0.05 cm)

Remember that the equation for the torque of a couple is the magnitude of only **one** of the forces multiplied by the distance between them, it's a common mistake to multiply both forces.

Question 10 Medium [1 mark]

A car wheel nut can be loosened by applying a force of 310 N to the end of a bar of length 0.7 m, as in **X**. A car mechanic is capable of applying forces of 650 N simultaneously in opposite directions on the ends of a wheel wrench, as in **Y**.



What is the minimum length l of the wrench needed for the mechanic to loosen the nut?

A 0.2 m

B 0.3 m

C 3.0 m

D 6.0 m

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The moment of a force is defined by:
  - Moment = Force × Perpendicular distance from the pivot
- The moment in **X** is therefore:
  - moment = $310 \times 0.7 = 217 \text{ N m}$
- This is the same moment that has to be applied in **Y**
- The torque of a couple is defined by:
  - Torque = Force × Perpendicular distance between the forces
- The torque of the couple is:
  - Torque = $650l = 217 \text{ N m}$
- Rearranging for length gives:

- $l = \frac{217}{650} = 0.33 \text{ m}$

- Therefore, $l = \frac{\text{moment X}}{\text{torque}} = \frac{310 \times 0.7}{650} = 0.33 \text{ m}$
    - This is answer **B**

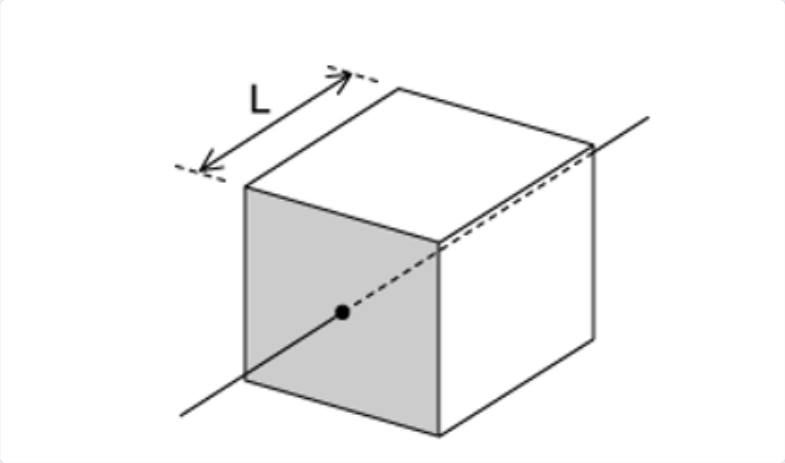
A is incorrect as the ratio has been calculated as $\frac{650}{217}$ which is the wrong way around

C is incorrect as 2 × 650 N has been used instead of just 650 N

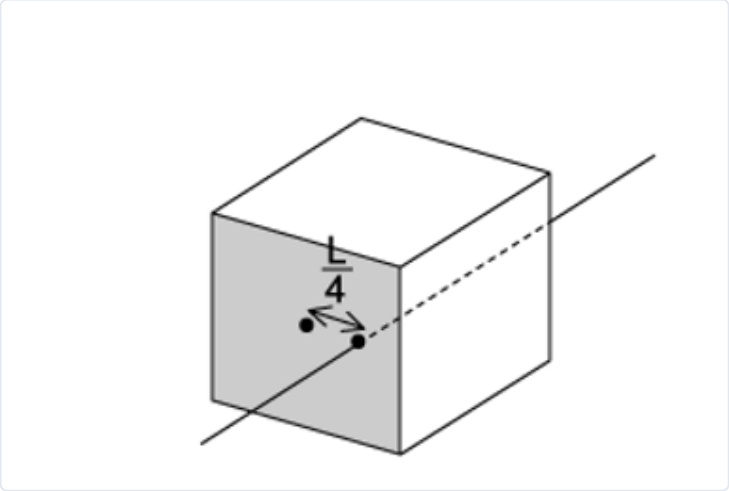
D is incorrect as the ratio has been calculated as $\frac{650}{217}$ which is the wrong way around and 2 × 650 N has been used instead of just 650 N

Question 11 Hard [1 mark]

The diagram shows a solid cube with weight *W* and sides of length *L*. It is supported by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.



When viewing the shaded face, what is the magnitude of the torque required to stop the cube from toppling?

A $\frac{WL}{2}$ anticlockwise

B $\frac{WL}{2}$ clockwise

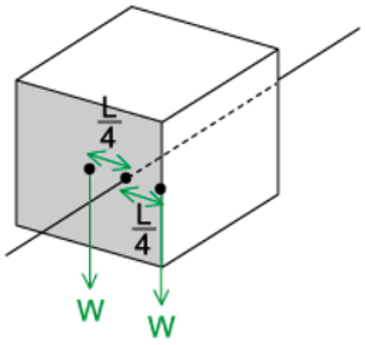
C $\frac{WL}{4}$ anticlockwise

D $\frac{WL}{4}$ clockwise

Mark Scheme & Explanation:

*The correct answer is **D** because:*

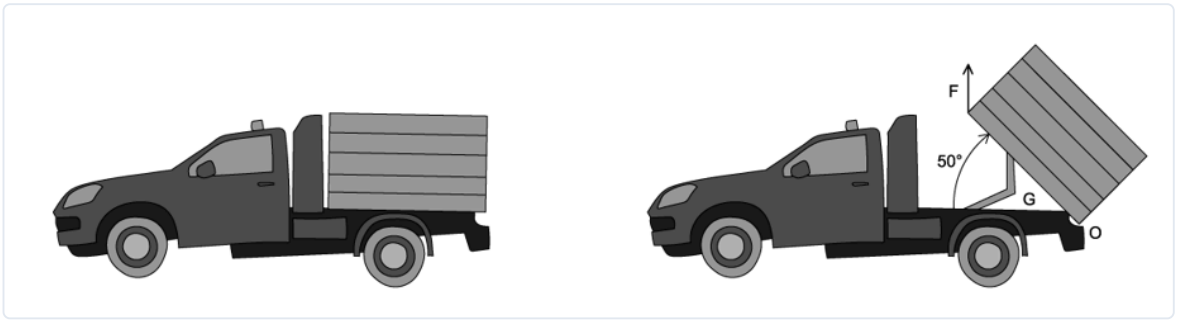
- We need to find the torque required to maintain equilibrium
    - This is not the torque produced by the centre of mass of the cube
- The weight acts downwards and is to the left of the pivot, therefore, this causes an anticlockwise moment at a perpendicular distance of $\frac{L}{4}$ from the spindle
    - Anticlockwise moment due to weight = $W \times \frac{L}{4} = \frac{WL}{4}$
- For equilibrium:
    - Sum of clockwise moments = Sum of anticlockwise moments
- In a couple, the perpendicular distance between the forces and the pivot (the spindle) should be equal
- So, to maintain equilibrium, a force equal to W should act along the vertical edge of the shaded face, which is a distance of $\frac{L}{4}$ to the right of the pivot
    - Clockwise moment to be applied = $W \times \frac{L}{4} = \frac{WL}{4}$
- Therefore, the torque of the couple that is required to prevent toppling is $\frac{WL}{4}$ clockwise



Question 12 Hard [1 mark]

A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50˚ to the horizontal by a vertical force *F* as shown in the diagram.

The trailer is 2.7 m in length, and the position of its centre of gravity G is a third of the length from the hinge at O.



What is the vertical force *F* applied to the end of the trailer?

A 10 kN

B 15 kN

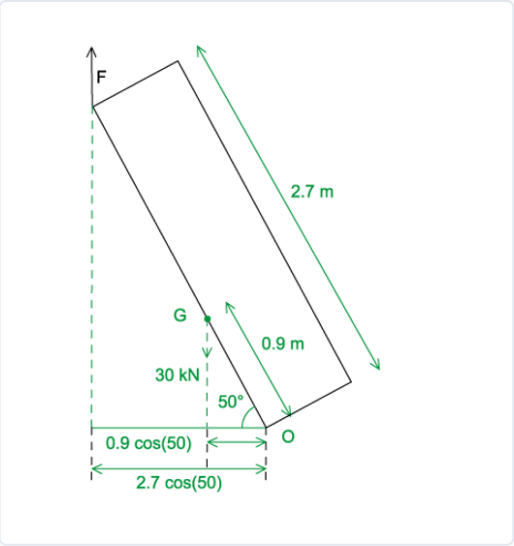
C 30 kN

D 60 kN

Mark Scheme & Explanation:

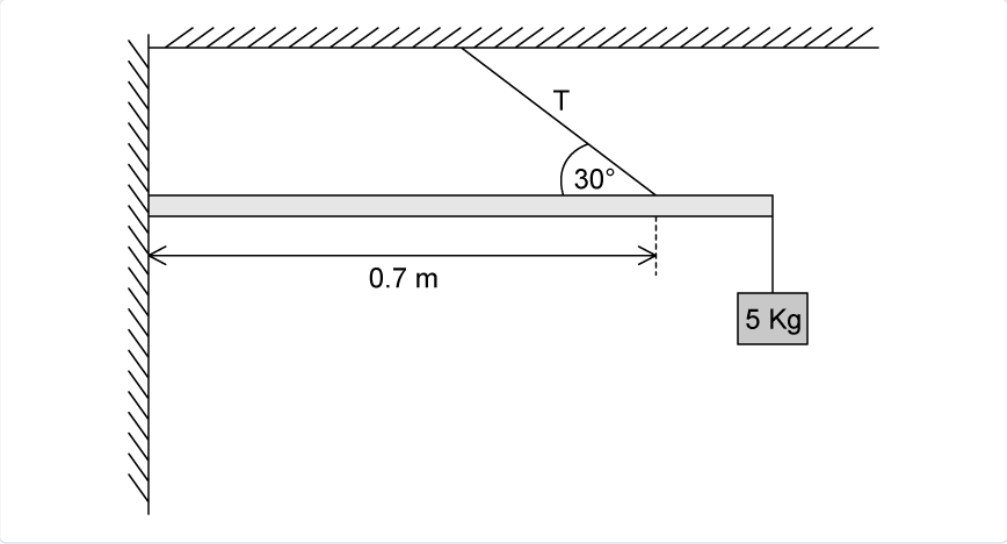
*The correct answer is **A** because:*

- The position of the trailer's centre of gravity G is a third of the length from the hinge at O
  - Therefore, the weight acts at a distance of $\frac{2.7}{3} = 0.9$ m from O
- Consider the moments around pivot O
  - The vertical force *F* provides a clockwise moment of $F \times 2.7 \cos 50°$
  - The weight of the trailer provides an anticlockwise moment of $30\ 000 \times 0.9 \cos 50°$
- For equilibrium, the clockwise moment should equal the anticlockwise moment
  - $F \times 2.7 \cos 50° = 30\ 000 \times 0.9 \cos 50°$
  - $F = \frac{30\ 000 \times 0.9 \cos 50°}{2.7 \cos 50°} = 10\ 000$ N
- The vertical force is therefore equal to 10 000 N or 10 kN


F
2.7 m
G
0.9 m
30 kN
50°
O
0.9 cos(50)
2.7 cos(50)

Question 13 Hard [1 mark]

A mass of 5 kg is attached to the end of a 1 m long uniform beam of mass 2 kg. The beam is fixed to a wall at one end and supported by a wire at an angle of 30° to the beam, 0.7 m from the wall. The beam is in equilibrium.



What is the tension *T* in the wire?

A 17 N

B 84 N

C 140 N

D 170 N

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- As the beam is uniform, its weight $mg = 2 \times 9.8$ acts at its centre of gravity at the midpoint, which is 0.5 m from the pivot
- The clockwise moment due to the beam's weight is:
  - $\text{moment} = \text{weight} \times \text{perpendicular distance from pivot}$
  - $\text{moment} = 2 \times 9.8 \times 0.5 = 9.8 \text{ N m}$
- The 5 kg mass is 1 m away from the pivot
- The clockwise moment due to the 5 kg mass is:
  - $\text{moment} = \text{mass} \times g \times \text{distance}$

- $\text{moment} = 5 \times 9.8 \times 1 = 49 \text{ N m}$

- The upward component of tension $T \sin 30°$ in the wire acts at a distance of 0.7 m from the pivot
- The anticlockwise moment due to the wire is:
  - $\text{moment} = 0.7 \times T \sin 30°$
- As the system is in equilibrium, the clockwise and anticlockwise moments must be equal:
  - $0.7 T \sin 30° = 9.8 + 49$
  - $T = \frac{58.8}{0.7 \sin 30°} = 168 \text{ N}$
- The tension in the wire is $T \approx 170 \text{ N}$ (to 2 s.f.)

A is incorrect because this is the answer if mass is used instead of weight in the calculation.

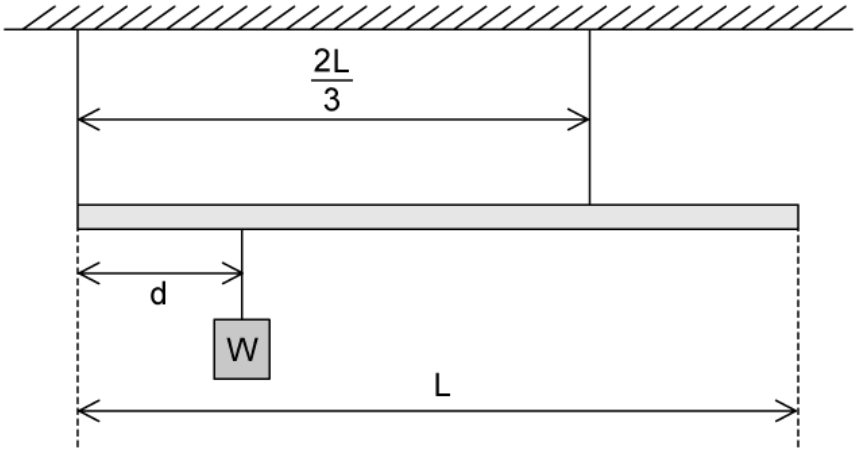
B is incorrect because this is the answer if the tension in the wire is not calculated from its vertical component.

C is incorrect because this is the answer if the mass of the beam is neglected from the calculations.

Question 14 Hard [1 mark]

A horizontal beam of length L and weight $\frac{W}{2}$ is suspended from the ceiling by two ropes.

One rope is attached at the left end of the beam, and the other is attached at a distance $\frac{2L}{3}$ from the left end. A load of weight W is suspended from the beam at a distance d from the left end, as shown. The beam is in equilibrium.



What value of d ensures that the tension is equal in both ropes?

A $\frac{L}{8}$

B $\frac{L}{4}$

C $\frac{L}{3}$

D $\frac{L}{2}$

Mark Scheme & Explanation:

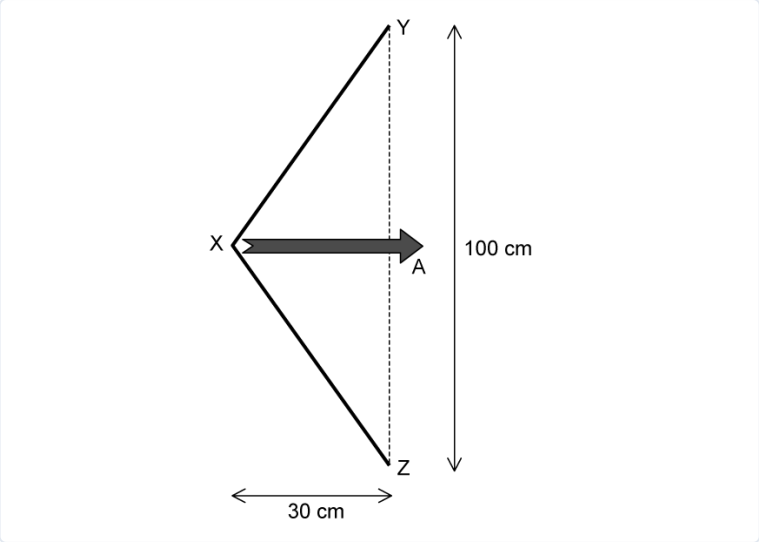
*The correct answer is **B** because:*

- If the tensions in each rope are equal, and the beam is in equilibrium:
  - Upwards forces = downward forces
  - $T + T = W + \frac{W}{2}$
  - $2T = \frac{3W}{2}$
  - $T = \frac{3W}{4}$
- Taking the left end of the beam as the pivot, the clockwise moment due to the load of weight W is:
  - $\text{moment} = W \times d$

- The beam is uniform, so its centre of gravity is halfway along the beam
- The clockwise moment due to the weight $\frac{W}{2}$ of the beam is:
  - $\text{moment} = \frac{W}{2} \times \frac{L}{2}$
  - $\text{moment} = \frac{WL}{4}$
- The tension T of the rope at a distance of $\frac{2L}{3}$ from the pivot produces an anticlockwise moment of:
  - $\text{moment} = T \times \frac{2L}{3}$
- For the beam to be in equilibrium, the anticlockwise and clockwise moments must be the same:
  - $T \times \frac{2L}{3} = Wd + \frac{WL}{4}$
- Substitute the expression for tension T and simplify:
  - $Wd = \frac{3W}{4}\frac{2L}{3} - \frac{WL}{4}$
  - $d = \frac{6L}{12} - \frac{L}{4}$
  - $d = \frac{L}{4}$

Question 15 Hard [1 mark]

The diagram shows a bow and arrow. The arrow is positioned at the centre X of the string of the bow stretched between points Y and Z. When the string is pulled back, the tension in the bowstring is 10 N.



What force will accelerate the arrow in the direction of point A when the string is released?

A 5.2 N

B 5.8 N

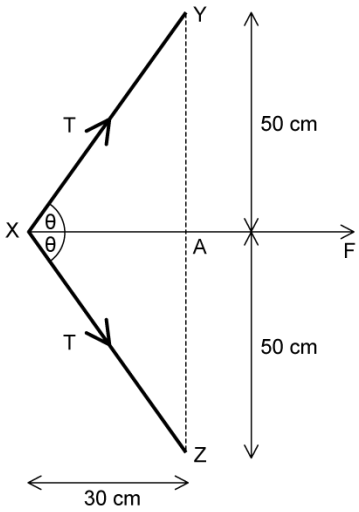
C 8.6 N

D 10.3 N

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Calculate the horizontal component of the tension of each side of the string.
  - $T \cos \theta$
  - Where T is the tension in the string, and θ is the angle between XY and XA.



- The bowstring is in equilibrium so the force, *F*, on the arrow from both halves of the string together will be:
  - $F = T \cos \theta + T \cos \theta = 2T \cos \theta$
- To find *θ*, we can use tan with adjacent side XA = 30 cm, and opposite side ZA = YA = 50 cm:
  - $\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{50}{30}$
  - $\theta = \tan^{-1} \frac{50}{30} = 59°$
- $F = 2T \cos \theta$
- $F = 2 \times 10 \times \cos 59° = 10.3 \text{ N}$

A is incorrect because this is the answer if only one half of the bowstring is included in the calculation

B is incorrect because this is the answer if opposite and adjacent lengths are accidentally swapped

C is incorrect because this is the answer the distance YZ is not halved when calculating the length of the opposite side of the triangle XYA